

Shoua J. Lee

Data Scientist | Decision Science | Applied AI

Milwaukee, WI | Willing to Relocate | (414) 745-1610 | sjslee557@gmail.com

GitHub: github.com/sjslee | Portfolio: <https://sjslee.github.io/Portfolio/>

Professional Summary

Analytical professional with experience spanning research, risk analytics, data science, and decision science across finance, banking, and marketing. Skilled in predictive modeling, experimentation, causal inference, forecasting, and translating complex analytical findings into actionable business decisions. Recently expanded expertise by developing AI-powered decision support applications that improve model interpretability and user experience.

Technical Skills

Programming: R, Python, SQL

Machine Learning: Classification, Regression, Time Series Forecasting

Econometrics/Decision Science: Uplift Modeling, Causal Inference, Experimentation, Driver Analysis

Generative AI: LLM Applications, Prompt Engineering, Ollama, OpenRouter

Data & Tools: Databricks (PySpark, Sparklyr), Snowflake, Streamlit, Hugging Face, Git, Docker, Excel, Qualtrics

Applied Data Science & AI Projects

Independent Projects | 2025–Present

AI Mortgage Credit Risk Assistant

Technologies: Python, Streamlit, Logistic Regression, OpenRouter

- Built and deployed an AI-powered mortgage credit risk assistant integrating a validated probability of default (PD) model with AI-generated explanations, interactive what-if scenario analysis, and decision support.

Mortgage Probability of Default model

Technologies: Python, R, Logistic Regression, WoE, Credit Risk Modeling

- Developed and validated a mortgage probability of default (PD) model using logistic regression and Weight of Evidence (WoE), achieving strong discrimination and stability through AUC, KS, calibration, PSI, and out-of-time validation.

Housing Affordability Forecasting Dashboard

Technologies: Python, ARIMA, FRED data, Zillow data

- Developed an interactive housing affordability dashboard integrating Zillow and FRED data with automated ARIMA forecasting to visualize affordability trends across U.S. states.

Data Analytic Consultant Senior

Associated Bank | Milwaukee, WI

June 2024 – Oct 2024

- Developed account-level interest rate elasticity models using Snowflake SQL to support deposit pricing strategies and evaluate customer sensitivity to interest rate changes.
- Analyzed large-scale banking account data in Snowflake to develop interest rate insights and support strategic decision-making.

Data Scientist

Leo Burnett Chicago | Chicago, IL

August 2021 – January 2024

- Conducted customer satisfaction analysis using **Key Driver Analysis** and **Penalty–Reward Analysis** to identify the primary drivers of customer satisfaction and support business decision-making.
- Developed a **coupon response model** that achieved approximately **3× lift** through targeted audience selection and decile validation.
- Built **uplift models** to measure the incremental impact of coupon redemption strategies and improve marketing campaign targeting.
- Forecasted website login traffic using time series forecasting to support resource planning and capacity management.

Research Analyst

Toastmasters International | Denver, CO

July 2019 – August 2021

- Designed and analyzed **controlled experiments (A/B tests)** using logistic and probit regression to evaluate email renewal strategies.
- Developed **Latent Class Analysis (LCA)** models to segment members into behavioral personas for targeted engagement strategies.
- Designed and analyzed **mixture design experiments** to optimize the "Success Recipe" program.
- Conducted customer satisfaction analysis using **Key Driver Analysis** and **Penalty–Reward Analysis** to identify factors influencing member experience.
- Built **contingent valuation** and **willingness-to-pay** models to evaluate online membership pricing strategies.

Risk Analyst: Quantitative Risk Modeling

Mortgage Guaranty Insurance Corporation (MGIC) | Milwaukee, WI

August 2017 – January 2018

- Conducted review, verification, and validation of the **Own Risk and Solvency Assessment (ORSA)** framework to support regulatory risk management and model governance.

Research Assistant

University of Wisconsin–Milwaukee | Milwaukee, WI

May 2014 – January 2017

- Applied quantitative and qualitative research methods, including survey design, interviews, and statistical analysis, to support academic research projects.

EDUCATION

University of Wisconsin Milwaukee

Masters in Economics (Applied Econometrics)

Bachelor in Economics, cum laude

Professional Activities & Awards

Instructor | Teach in Thailand | 2025

- Designed and taught English curriculum for 500+ students.

Awards & Research Grants

- Awarded competitive research grants to conduct international field research in **French Guiana (Strategic Marketing, 2015)** and **Argentina (2016)**.